Simple random sampling with replacement (SRSWR):

If the selected cards are replaced before the next draw, such a sampling is called sampling with replacement

Remark:

If the population size is large, this method is cumbersome. The alternative method is using of table of random numbers.

(2) Table of Random numbers

The easiest way to select a sample randomly is to use random number tables. There are several tables of random numbers. Some of them are

(i) Kendall and Smith random number table.

(ii) Tippet's random number table.

(iii) Fishers and Yates random number table.

Method of using Random Number table.

In a given finite population, the method of selecting a random sample from a random number table is given below:

The table contains different numbers consisting of 0, 1, 2,… ,9 and possesses the essential characteristics which ensures random sampling. A part of the table is given as Appendix. Referring to this page, selection of a random sample is explained by taking a finite population of size 100 units and selecting a sample of 10 units. The steps to be followed are listed as under.

(i) A list of all 100 units in the finite population is prepared and each unit is assigned a serial number ranging from 00 to 99.

(ii) Any number in the table is chosen at random and it is the starting point for selecting the sampling units.

(iii) From the starting point we can make a move on to the next number either vertically, horizontally or diagonally.

(iv) Since our numbered population consists of two digits from 00 to 99 , we confine ourselves to reading only two digit numbers without omitting any number that comes forward.

(v) As we proceed, random numbers read from the table which are above 99 are ignored and those numbers which are less than are equal to 99 are recorded. This process is continued until we reach 10 such random numbers.

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